Homework 1

Austin Bennett

Q.1) Why is main memory unsuitable for permanent program storage or backup purposes? Furthermore, what is the main disadvantage to storing information on a magnetic disk drive as opposed to main memory?

Answer: The main reason is capacity RAM is smaller than disk drives. The problem with this is that RAM is typically faster than a drive.

Q.2) Answer the following questions regarding interrupts in an operating system.

a. What is the purpose of the interrupt vector? In other words, what advantage does it provide over a general-purpose interrupt-processing routine?
b. How does an interrupt differ from a trap?
c. Can traps be generated intentionally by a user program? If so, for what purpose?

Answer:
A: It allows the OS to respond to hardware interrupts. It can be more efficient and organized.
B: An interrupt are events made by outside eactions that are outside the executing program. A trap is inside the program.
C: Yes, typically as a system call like reading a file.

Q.3) Why is multi-core processing more efficient than placing each processor on its own chip?

Answer: Communication between cores is easier

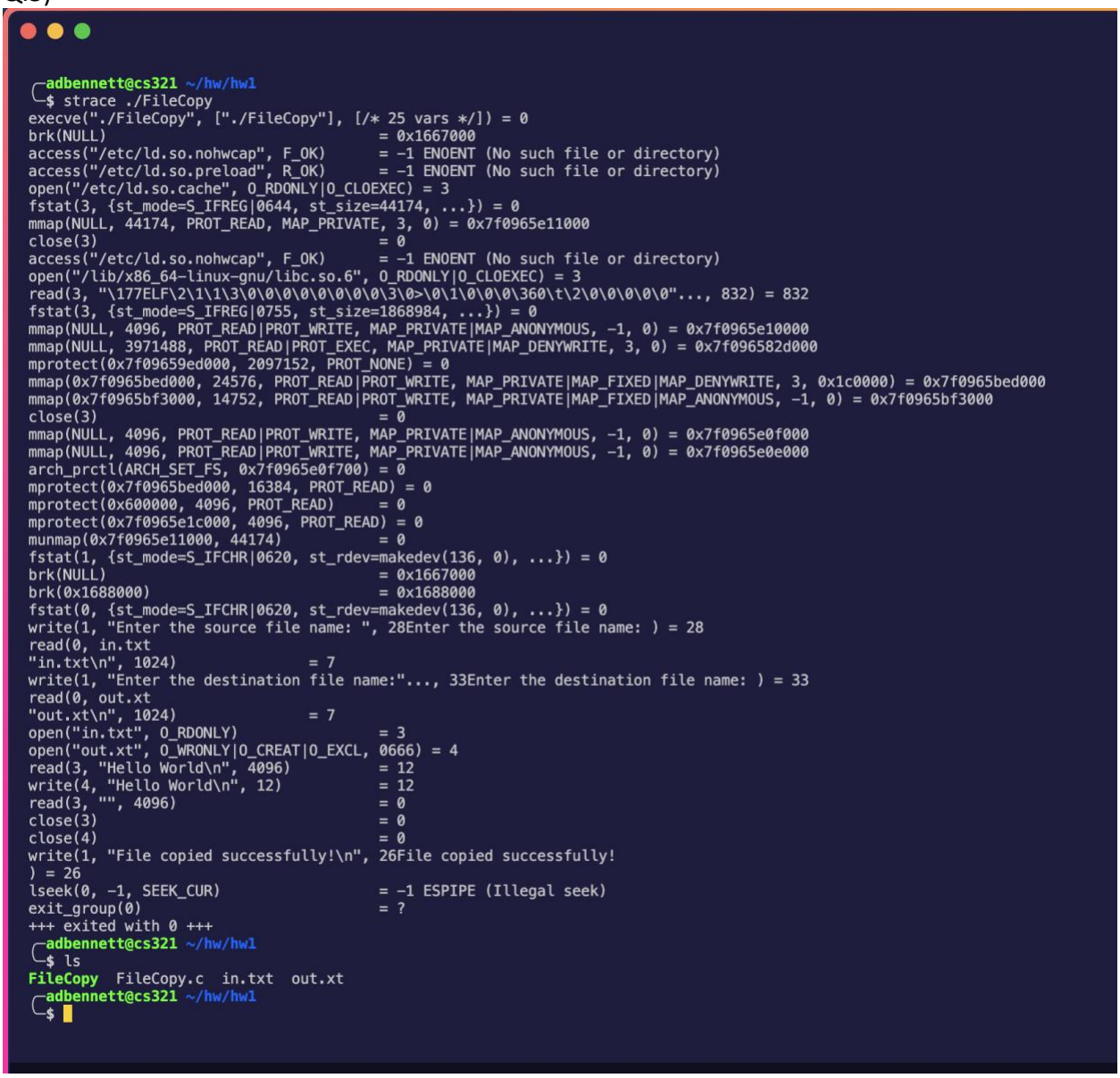
Q.4) Suppose we want to start a new process on a UNIX system and load a program into it. Which system calls must be executed by a command interpreter or shell to achieve that?

Answer: fork() exec()

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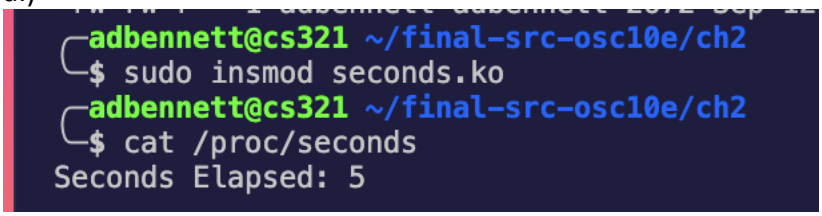
Q.5)



```
adbennett@cs321 ~/hw/hw1
$ strace ./FileCopy
execve("./FileCopy", ["./FileCopy"], [/* 25 vars */]) = 0
brk(NULL)                               = 0x1667000
access("/etc/ld.so.nohwcap", F_OK)      = -1 ENOENT (No such file or directory)
access("/etc/ld.so.preload", R_OK)      = -1 ENOENT (No such file or directory)
open("/etc/ld.so.cache", O_RDONLY|O_CLOEXEC) = 3
fstat(3, {st_mode=S_IFREG|0644, st_size=44174, ...}) = 0
mmap(NULL, 44174, PROT_READ, MAP_PRIVATE, 3, 0) = 0x7f0965e11000
close(3)                                = 0
access("/etc/ld.so.nohwcap", F_OK)      = -1 ENOENT (No such file or directory)
open("/lib/x86_64-linux-gnu/libc.so.6", O_RDONLY|O_CLOEXEC) = 3
read(3, "\177ELF\2\1\1\3\0\0\0\0\0\0\0\0\3\0>\0\1\0\0\0\360\t\2\0\0\0\0\0"..., 832) = 832
fstat(3, {st_mode=S_IFREG|0755, st_size=1868984, ...}) = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f0965e10000
mmap(NULL, 3971488, PROT_READ|PROT_EXEC, MAP_PRIVATE|MAP_DENYWRITE, 3, 0) = 0x7f096582d000
mprotect(0x7f09659ed000, 2097152, PROT_NONE) = 0
mmap(0x7f0965bed000, 24576, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_DENYWRITE, 3, 0x1c0000) = 0x7f0965bed000
mmap(0x7f0965bf3000, 14752, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_FIXED|MAP_ANONYMOUS, -1, 0) = 0x7f0965bf3000
close(3)                                = 0
mmap(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f0965e0f000
mmap(NULL, 4096, PROT_READ|PROT_WRITE, MAP_PRIVATE|MAP_ANONYMOUS, -1, 0) = 0x7f0965e0e000
arch_prctl(ARCH_SET_FS, 0x7f0965e0f700) = 0
mprotect(0x7f0965bed000, 16384, PROT_READ) = 0
mprotect(0x600000, 4096, PROT_READ)     = 0
mprotect(0x7f0965e1c000, 4096, PROT_READ) = 0
munmap(0x7f0965e11000, 44174)           = 0
fstat(1, {st_mode=S_IFCHR|0620, st_rdev=makedev(136, 0), ...}) = 0
brk(NULL)                               = 0x1667000
brk(0x1688000)                          = 0x1688000
fstat(0, {st_mode=S_IFCHR|0620, st_rdev=makedev(136, 0), ...}) = 0
write(1, "Enter the source file name: ", 28Enter the source file name: ) = 28
read(0, in.txt
"in.txt\n", 1024)              = 7
write(1, "Enter the destination file name:"..., 33Enter the destination file name: ) = 33
read(0, out.xt
"out.xt\n", 1024)              = 7
open("in.txt", O_RDONLY)                = 3
open("out.xt", O_WRONLY|O_CREAT|O_EXCL, 0666) = 4
read(3, "Hello World\n", 4096)          = 12
write(4, "Hello World\n", 12)           = 12
read(3, "", 4096)                       = 0
close(3)                                = 0
close(4)                                = 0
write(1, "File copied successfully!\n", 26File copied successfully!
) = 26
lseek(0, -1, SEEK_CUR)                  = -1 ESPIPE (Illegal seek)
exit_group(0)                           = ?
+++ exited with 0 +++
adbennett@cs321 ~/hw/hw1
$ ls
FileCopy  FileCopy.c  in.txt  out.xt
adbennett@cs321 ~/hw/hw1
$
```

Q.6)

a.)



```
adbennett@cs321 ~/final-src-osc10e/ch2
$ sudo insmod seconds.ko
adbennett@cs321 ~/final-src-osc10e/ch2
$ cat /proc/seconds
Seconds Elapsed: 5
```

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b.)

